Introduction to Mathematics for Deep Learning: Homeworks #2 Solutions

1. **Maximum Differential Entropy.** Consider a random variable $X \in [a, b]$ with probability 1, what is the maximum differential entropy?

   Solution: Maximum Differential Entropy.

$$
\begin{aligned}
h(X) =& \mathbb{E}\left[\log \tfrac{1}{f_X(x)}\right] && (1)\\
=& \mathbb{E}\left[\log \tfrac{1}{f_X(x)}\right] && (2)\\
\geq& \log \mathbb{E}\left[\frac{1}{f_X(x)}\right] && (3)\\
=& \log \int_a^b \frac{1}{f_X(x)} f_X(x)\, dx && (4)\\
=& \log(b-a). && (5)
\end{aligned}
$$

2. **Gaussian random variables**
   Let X be a Gaussian random variable (r.v) with mean 0 and variance 1. Set

$$
Y = \begin{cases} -X & \text{if } |X| \leq c, \\ X & \text{if } |X| > c. \end{cases}
$$

   (a) What is the distribution of Y ?

   (b) What is the joint distribution of (X, Y)? Is it a multivariate Gaussian distribution?

   (c) Can you find a value of c such that $\mathbb{E}[XY] = 0$ (not necessary in closed form)? For this c value, is Y independent of X?

   (d) For $n > 2$, can you find a random vector $(X_1, \cdots, X_n)$ such that the marginal distribution for any m $(m < n)$ of them is a m dimensional Gaussian distribution, but the joint distribution for $(X_1, \cdots, X_n)$ is not a n dimensional Gaussian distribution?

   Solution: Gaussian random variables

   (a) Y has a same marginal distribution with X, which is $\mathcal{N}(0, 1)$.

   (b) It is not a multivariate Gaussian distribution.
   Suppose (X, Y) are multivariate Guassian. $P(|X| = |Y|) = 1$ implies that the correlation ρ should be either -1 or 1. However, which also means $Y = \rho X$, and this is not the case.

(c) Let f_X be pdf of X, which is standard normal random variable.

$$\mathbb{E}[XY] = \int_{-\infty}^{-c} x^2 f_X(x)dx + \int_{-c}^{c} -x^2 f_X(x)dx + \int_{c}^{\infty} x^2 f_X(x)dx$$

and we can find a such c that makes $\mathbb{E}[XY]$.

Since (X, Y) are not multivariate Gaussian, $\mathbb{E}[XY] = 0$ does not implies the independence. Note that the marginal distribution of Y is normal but Y is deterministic conditioned on X. Therefore, X and Y are not independent.

(d) We can construct the such random vector in the following manner. Let $X_1, \cdots, X_{n-1}$ and Z be i.i.d. Gaussian random variables with mean 0 and variance 1. Define $\text{sgn}(x)$ be a sign of x (either 1 or -1), and then let the random variable X_n be

$$X_n = |Z| \times \prod_{i=1}^{n-1} \text{sgn}(X_i).$$

Similar to (a), X_n is Gaussian random variable with mean 0 and variance 1. However, sign of X_n is determined by $(X_1, \cdots, X_{n-1})$ which implies that the joint distribution for $(X_1, \cdots, X_n)$ is not a n dimensional Gaussian distribution.

On the other hand, for any $m < n$ and indices $1 \leq i_1 < \cdots < i_m \leq n$, $(X_{i_1}, \cdots, X_{i_m})$ are jointly Gaussian. If $i_m < n$, then $(X_{i_1}, \cdots, X_{i_m})$ is clearly m dimensional i.i.d. Gaussian random variables. If $i_m = n$, X_n and $(X_{i_1}, \cdots, X_{i_{m-1}})$ are indepedent as well. This is mainly because the sign of X_n can not be determined without full sign information of $(X_1, \cdots, X_{n-1})$.

3. **Differential entropy.**
Evaluate the differential entropy $h(X) = -\int f \ln f$ for the following:

(a) The exponential density, $f(x) = \lambda e^{-\lambda x}$, $x \geq 0$.

(b) The Laplace density, $f(x) = \frac{1}{2}\lambda e^{-\lambda|x|}$.

(c) The sum of X_1 and X_2, where X_1 and X_2 are independent normal random variables with means μ_i and variances $\sigma_i^2, i = 1, 2$.

Solution: Differential Entropy.

(a) Exponential distribution.

$$h(f) = -\int_0^{\infty} \lambda e^{-\lambda x}[\ln \lambda - \lambda x]dx \tag{6}$$

$$= -\ln \lambda + 1 \text{ nats.} \tag{7}$$

$$= \log \frac{e}{\lambda} \text{ bits.} \tag{8}$$

(b) Laplace density.

$$h(f) = -\int_{-\infty}^{\infty} \frac{1}{2}\lambda e^{-\lambda|x|}[\ln\frac{1}{2} + \ln\lambda - \lambda|x|]\,dx \tag{9}$$

$$= -\ln\frac{1}{2} - \ln\lambda + 1 \tag{10}$$

$$= \ln\frac{2e}{\lambda} \text{ nats.} \tag{11}$$

$$= \log\frac{2e}{\lambda} \text{ bits.} \tag{12}$$

(c) Sum of two normal distributions.

The sum of two normal random variables is also normal, so applying the result derived the class for the normal distribution, since $X_1 + X_2 \sim \mathcal{N}(\mu_1 + \mu_2, \sigma_1^2 + \sigma_2^2)$,

$$h(f) = \frac{1}{2}\log 2\pi e(\sigma_1^2 + \sigma_2^2) \text{ bits.} \tag{13}$$

4. **Markov's inequality for probabilities.**
Let $p_X(x)$ be a probability mass function of random variable X. Prove, for all $d \geq 0$,

$$P(p_X(X) \leq d)\ \log\left(\frac{1}{d}\right) \leq H(X).$$

Solution: Markov inequality applied to entropy.

$$P(p_X(X) < d)\log\frac{1}{d} = \sum_{x:p_X(x)<d} p_X(x)\log\frac{1}{d} \tag{14}$$

$$\leq \sum_{x:p_X(x)<d} p_X(x)\log\frac{1}{p_X(x)} \tag{15}$$

$$\leq \sum_{x} p_X(x)\log\frac{1}{p_X(x)} \tag{16}$$

$$= H(X) \tag{17}$$

5. **MLE** Let $X_1, \ldots, X_n \sim p_X$ be Poisson random variable with parameter λ. I.e.,

$$p_X(k) = \frac{\lambda^k e^{-\lambda}}{k!}. \tag{18}$$

(a) Find a MLE, i.e., $\hat{\lambda}_{MLE}$.

(b) Is $\hat{\lambda}_{MLE}$ unbiased?

(c) Find a mean squred error of $\hat{\lambda}_{MLE}$ as a function of λ.

(a) MLE is to maximize the log likelihood, i.e.,

$$\sum_{i=1}^{n} \log p_X(x_i) = \sum_{i=1}^{n} x_i \log \lambda - \lambda - \log x_i!. \tag{19}$$

If we take the derivative with respect to λ, we have

$$\sum_{i=1}^{n} \frac{x_i}{\lambda} - 1 = 0. \tag{20}$$

I.e.,

$$\hat{\lambda}_{MLE} = \frac{1}{n} \sum_{i=1}^{n} x_i. \tag{21}$$

(b) Since $\mathbb{E}[X_i] = \lambda$, we have

$$\mathbb{E}\left[\hat{\lambda}_{MLE}\right] = \lambda \tag{22}$$

which implies unbiasedness.

(c) The MSE is given by

$$\begin{aligned}
\mathbb{E}\left[(\hat{\lambda}_{MLE} - \lambda)^2\right] =& \frac{1}{n^2} \mathbb{E}\left[\sum_{i=1}^{n} (X_i - \lambda)^2\right] && (23)\\
=& \frac{1}{n^2} \sum_{i=1}^{n} \mathbb{E}\left[(X_i - \lambda)^2\right] && (24)\\
=& \frac{1}{n^2} \sum_{i=1}^{n} \lambda. && (25)\\
=& \frac{\lambda}{n} && (26)
\end{aligned}$$

where (24) is due to independence of X_i's and (25) is because the variance of Poisson distribution is λ.

6. Show that the uniform distribution is a stationary initial distribution if the transition probability matrix P for Markov process is symmetric.

Solution:

Let P be a $d \times d$ transition probability matrix where $P^T = P$. Then, for $\pi_u = \begin{bmatrix} 1/d & 1/d & \dots & 1/d \end{bmatrix}^T$, we want to show that $P\pi_u = \pi_u$. For any $1 \le j \le d$,

$$\sum_{i=1}^{d} P_{i,j} \pi_u(j) = \frac{1}{d} \sum_{i=1}^{d} P_{j,i} = \frac{1}{d} = \pi_u(i). \tag{27}$$

This concludes the proof.